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Computational study of defect engineered K_2GeI_6 based perovskite solar cell using multiscale modeling and machine learning

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Abstract

K_2GeI_6 is introduced as a novel, lead-free halide double perovskite designed to address the increasing demand for stable and eco-friendly photovoltaic technologies. K_2GeI_6 exhibits favourable electronic and optical properties as confirmed through DFT-based analyses, which highlight its suitable bandgap, strong absorption, and favourable other optical properties. Using a multiscale optimization strategy that integrates SCAPS-1D simulations with machine-learning-assisted defect engineering, we refined the FTO/WS_2/K_2GeI_6/MoO_3/Au architecture to suppress recombination losses and enhance carrier transport. SCAPS-1D results validate the strong optoelectronic performance of the absorber, while ML-guided defect introduction proves lesser degradation even at elevated interface defect densities, demonstrating remarkable defect resilience and operational robustness under real environmental conditions. With an excellent PCE of 27.83%, FF of 85.78%, Jsc of 30.58 mA/cm^2, and Voc of 0.98 V, the device establishes K_2GeI_6 as a promising lead-free candidate for next-generation high-performance photovoltaic applications, where defect tolerance, thermal stability, and long-term operational reliability are critical. The novelty of this work lies in the synergistic integration of DFT insights, SCAPS-1D device optimization, and ML-driven defect engineering to deliver a comprehensive computational framework for designing defect-resilient K_2GeI_6 perovskite solar cells.

Keywords SCAPS 1D, K_2GeI_6, Perovskite solar cells, Defect engineering, Machine learning

1 Introduction

The ever-growing global demand for clean and renewable energy has positioned solar photovoltaics (PV) as one of the most promising technologies to replace conventional fossil fuels [1]. By 2022, the installed PV capacity worldwide had already surpassed 1 TW, underlining the rapid adoption of solar energy [2]. While crystalline silicon solar cells dominate the market with record efficiencies above 26%, their high manufacturing cost, energy-intensive processing, and rigidity pose limitations for widespread and flexible applications [3]. This has encouraged researchers to explore new classes of low-cost, solution-processable, and high-efficiency materials that can complement or even surpass the performance of silicon-based devices [4].

Perovskite solar cells (PSCs) have emerged as strong contenders, achieving rapid efficiency improvements from 3.8% in 2009 to over 25% in just a decade [5, 6]. Recent theoretical investigations on oxide and double perovskites have demonstrated their significant potential for advanced optoelectronic, thermoelectric, and spintronic applications. Studies on $YGaO_3$ revealed structural stability, direct bandgap semiconducting behaviour, strong optical absorption, and ductile characteristics suitable for photovoltaic devices [7]. Similarly, $K_2ScAuCl_6$, $K_2ScAuBr_6$, Tl_2WCl_6, and Tl_2WBr_6 exhibited promising electronic, thermoelectric, and magnetic properties, including half-metallicity and high spin polarization, highlighting their suitability for multifunctional energy and spintronic technologies [8, 9].

Rb-based perovskites have been explored to stabilize the crystal structure and enhance efficiency, with reported PCEs exceeding 21% [10]. However, they suffer from poor moisture stability, limiting their operational lifetime. Similarly, Cs-based all-inorganic perovskites have demonstrated good thermal tolerance and efficiencies around 18–19%, but they encounter serious phase instability at room temperature, which degrades performance under prolonged illumination [11]. Sn-based perovskites were introduced as lead-free alternatives, with devices reaching efficiencies close to 14–15%. Unfortunately, their performance rapidly decays due to the oxidation of Sn^{2+} to Sn^{4+}, resulting in severe instability [12]. Despite these advances, a key limitation across these systems is the lack of emphasis on defect engineering, the presence of deep-level defects and trap states accelerates non-radiative recombination, significantly restricting efficiency and stability. Thus, addressing defects is critical for advancing next-generation PSCs.

To bridge these challenges, this work investigates K_2GeI_6 as a new halide double perovskite absorber with enhanced stability and defect tolerance. Density Functional Theory (DFT) analysis confirms that K_2GeI_6 possesses a direct bandgap and Density of states (DOS). The calculated dielectric constant, absorption coefficient, and other optical properties demonstrate excellent light-harvesting capability in the visible region, making K_2GeI_6 highly suitable for PV technologies and particularly promising for next-generation high-efficiency photovoltaic applications, where materials must demonstrate defect tolerance, thermal stability, and long-term operational reliability under standard operating conditions. Our optimized device configuration—FTO/WS_2/K_2GeI_6/MoO_3/Au—achieved outstanding performance with a PCE of 27.83%, fill factor (FF) of 85.78%, short-circuit current density (Jsc) of 30.58 mA/cm^2, and open-circuit voltage (Voc) of 0.98 V.

To fully understand and optimize device behaviour, a comprehensive multiscale framework was employed. SCAPS-1D simulations were carried out to tune layer thicknesses, doping densities, and energy-level alignment to ensure efficient charge transport and minimized recombination. In parallel, machine-learning-driven defect engineering was introduced to analyse and predict critical defect parameters. By identifying correlations between these microscopic defect states and macroscopic PV parameters, the ML model isolated the most detrimental defects and proposed defect passivation strategies.

The optimized ML-derived parameters were re-implemented into SCAPS-1D, validating the predictions and demonstrating significant improvements in stability and recombination suppression. This iterative SCAPS–ML feedback loop, supported by the underlying DFT-derived electronic and optical insights, effectively minimized non-radiative recombination, enhanced carrier lifetime, and improved interfacial charge extraction. The novelty of this approach lies in the synergistic integration of DFT, SCAPS-1D, and ML, creating a generalizable pathway for data-guided defect engineering in emerging lead-free perovskites. Furthermore, the superior defect resilience and thermal stability of K_2GeI_6 make it a strong candidate for durable next-generation PV modules, particularly for next-generation durable photovoltaic modules, where defect tolerance, thermal stability, and long-term operational reliability are essential. Recent experimental studies on germanium-based perovskites have demonstrated promising optoelectronic behavior, although practical device efficiencies remain limited by instability, defect-assisted recombination, and interface-related losses [13]. The comparatively high efficiency obtained in the present study is primarily attributed to the optimized and idealized simulation conditions employed in the SCAPS-1D model, including low defect density, suppressed recombination pathways, and efficient charge transport. Therefore, the obtained power conversion efficiency represents a theoretical upper-performance limit under ideal operating conditions.

The novelty of the present work lies not only in the optimization of device parameters using the SCAPS-1D framework, but also in the systematic investigation of the comparatively less explored lead-free K_2GeI_6 absorber material integrated with machine learning-assisted performance analysis. Unlike conventional SCAPS studies focused mainly on parameter variation, this work establishes correlations between critical material/device parameters and photovoltaic output characteristics to better understand the governing physical mechanisms affecting device efficiency. Furthermore, the study provides a detailed assessment of defect-assisted recombination, interface optimization, and charge transport behavior in K_2GeI_6-based solar cells, thereby contributing both predictive and mechanistic insights for the development of environmentally benign perovskite photovoltaics.

Figure 1(a) shows the schematic design of the K_2GeI_6-based perovskite solar cell with its layered structure, including FTO as the front contact, WS_2 as the electron transport layer, K_2GeI_6 as the absorber, MoO_3 as the hole transport layer, and Au as the back contact. Figure 1(b) presents the crystal structure of K_2GeI_6, where potassium (green), germanium (red), and iodine (blue) atoms are arranged in a stable double perovskite lattice. The highly symmetric octahedral coordination of GeI_6 units ensures structural stability and mitigates the ion migration issues commonly found in hybrid lead halide perovskites. Potassium ions occupy the interstitial sites, providing charge balance and enhancing lattice rigidity, which reduces defect formation and phase instability. The Ge–I framework plays a crucial role in tuning the bandgap and optical absorption properties, making K_2GeI_6 highly suitable as a lead-free light absorber. By suppressing deep-level defect states and enhancing carrier mobility, this structure directly contributes to improved efficiency, reduced recombination losses, and enhanced long-term stability of the solar cell. Figure 1(c) shows the comprehensive work done this study.

2 Simulation methodologies and input parameters

The DFT calculations are done by Cambridge Sequential Total Energy Package (CASTEP) module in materials studio. The input parameters were taken from previous works. The same are provided into Sect. 1.1 of supporting information.

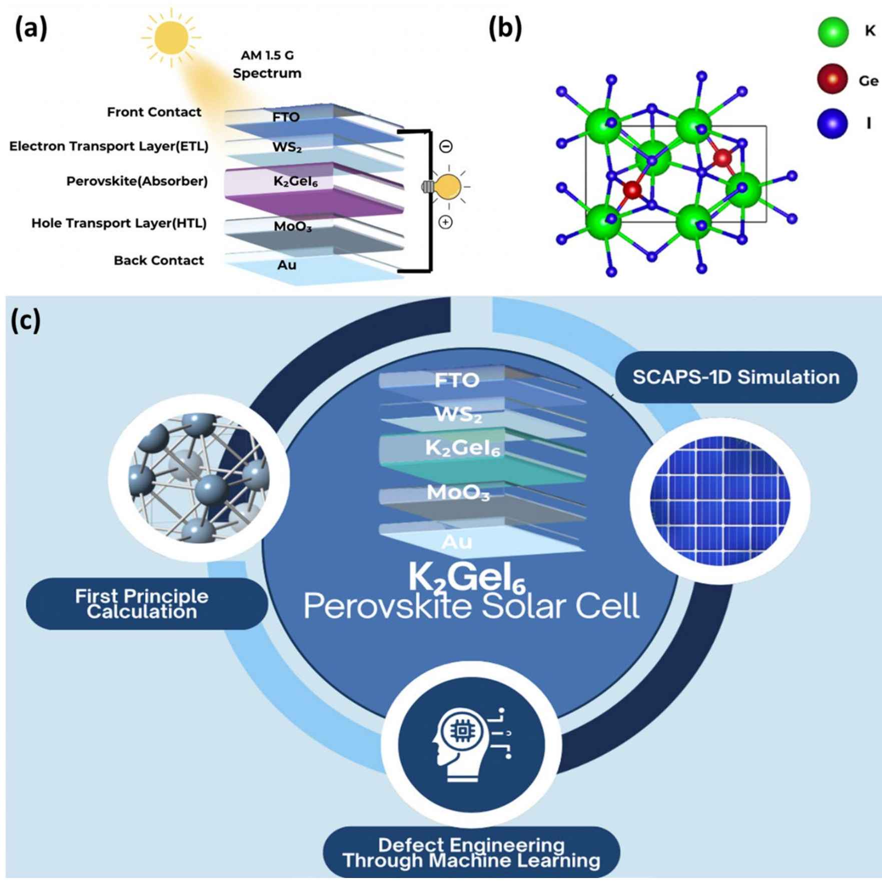

Fig. 1 (**a**) Proposed solar cell architecture, (**b**) 3D schematic of K_2GeI_6 and (**c**) representation of this study

Device-level simulations were carried out using the Solar Cell Capacitance Simulator (SCAPS-1D) to analyze the performance of the proposed architecture FTO/WS_2/K_2GeI_6/MoO_3/Au [14]. The input parameters for each layer were taken from Table S1 (Supporting information). The detailed SCAPS-1D input parameters, including bandgap, electron affinity, carrier mobility, defect density, doping concentration, and dielectric constant values for each layer, are summarized in Supporting Information (Table S1). All simulation parameters were adopted from previously reported literature unless otherwise specified. The absorber bandgap value employed in this study was obtained from the present investigation and is explicitly indicated in Table S1. To complement SCAPS-1D simulations, a random forest regression model was employed for defect engineering in K_2GeI_6-based solar cell [15]. The dataset consisted of approximately 3520 samples compiled from previously reported simulation and experimental studies on K_2GeI_6 and similar perovskite configurations, consisting of defect-related and material-dependent parameters along with corresponding photovoltaic metrics [16, 17]. Key input features included bandgap energy (E_g), electron affinity (χ), dielectric constant (ε_r), electron mobility (μ_e), hole mobility (μ_h), and defect density (N_t), while the target output parameter was the PCE. Dataset was divided into the ratio of 80:20 for training and testing respectively [18]. The model demonstrated strong predictive capability, achieving an R^2 of 0.976, with RMSE of 0.27 and MAE of 0.20, highlighting excellent agreement between predicted and simulated PV metrics.

To evaluate the robustness and generalization capability of the Random Forest regression model, a 5-fold cross-validation strategy was employed during model training. The cross-validation results demonstrated stable predictive behavior with only minor variation across folds, confirming good model consistency and reduced risk of overfitting. The model achieved comparable performance for both training and testing datasets, with training and testing accuracies remaining closely aligned, indicating strong generalization capability.

The Random Forest model was implemented using commonly optimized hyperparameters, including 200 decision trees ($n_{estimators}$ = 200), maximum tree depth of 20 (max-depth = 20), minimum samples split of 2, minimum samples leaf of 1, and bootstrap sampling enabled. These settings provided a suitable balance between prediction accuracy and model complexity for the multidimensional photovoltaic dataset.

Feature importance analysis revealed that interface N_t, absorber thickness, carrier mobility, and doping concentration were among the most influential parameters affecting photovoltaic performance, particularly PCE. This analysis further confirms the dominant role of defect-assisted recombination and charge transport in governing device behavior. Figure 2 represents the ML workflow of this work.

3 Results and insights

3.1 DFT calculations-electronic and optical results

Figure 3(a) presents the electronic band structure of K_2GeI_6, revealing a direct bandgap of 1.28 eV, which is in close agreement with previously published value [19]. The smooth dispersion of the conduction and valence bands indicates relatively low effective masses, favouring efficient charge transport for a solar cell. Figure 3(b) shows the DOS, where strong peaks between – 2 and 0 eV originate mainly from I-5p and Ge-4p hybridization,

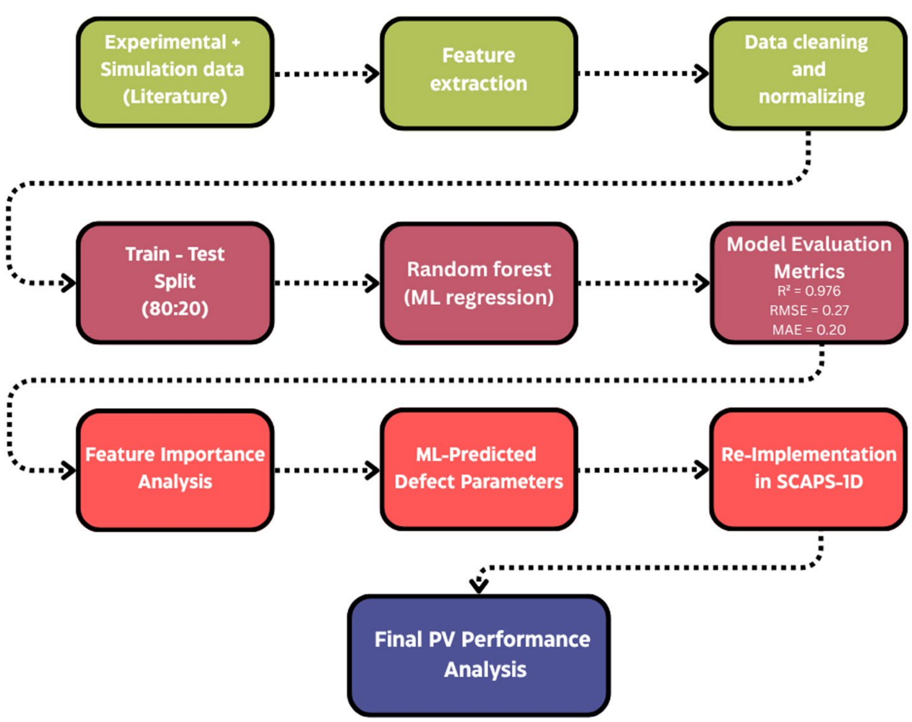

Fig. 2 ML workflow of this study

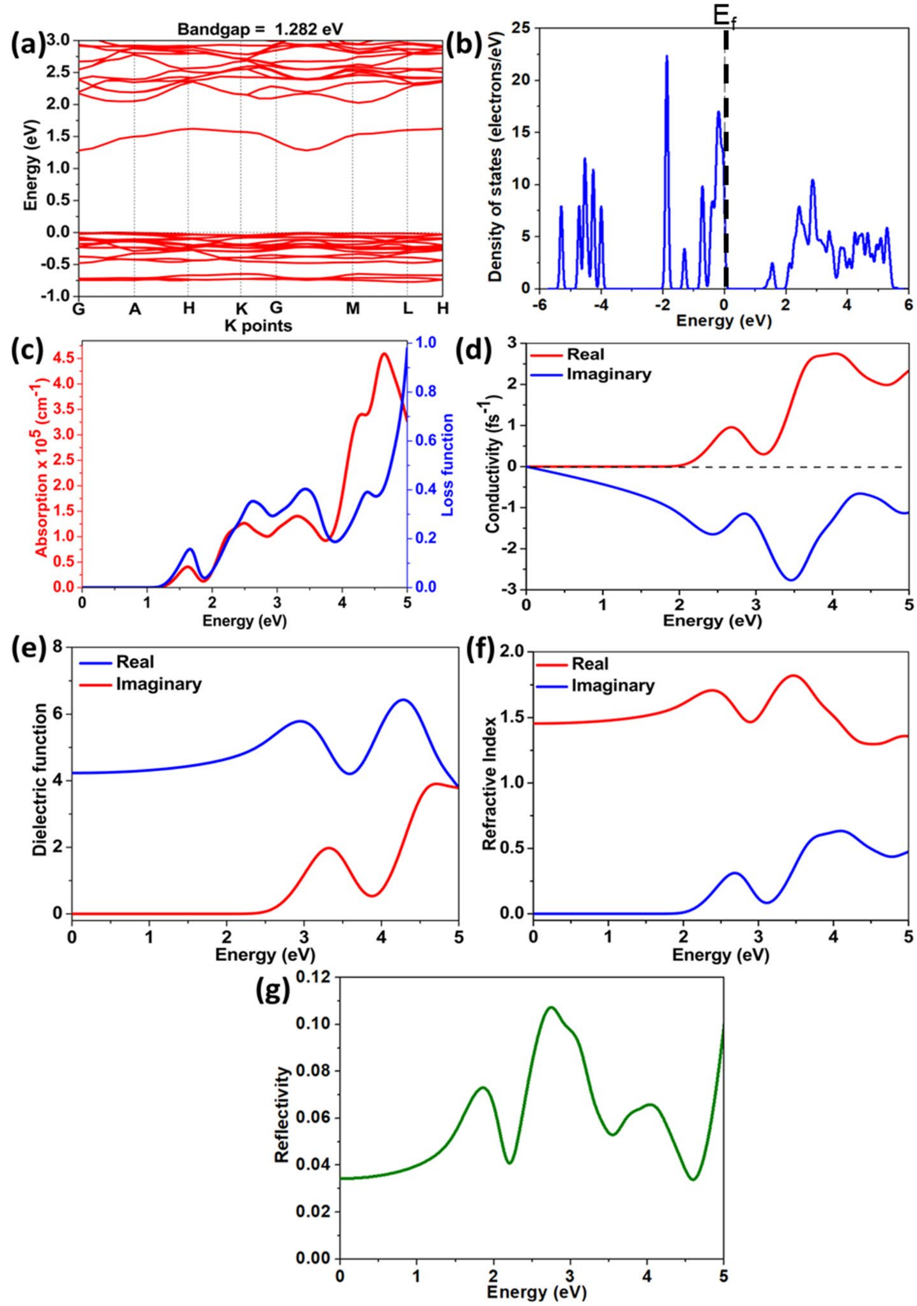

Fig. 3 (**a**) Bandgap, (**b**) DOS, (**c**) absorption and loss function, (**d**) conductivity, (**e**) dielectric function, (**f**) refractive index and (**g**) reflectivity

while conduction states above 1 eV are dominated by Ge-s orbitals. This indicates a well-bonded halide-germanium network that naturally suppresses deep defect states—highly beneficial for long-term structural and thermal stability under operational photovoltaic conditions.

Figure 3(c) displays the absorption coefficient and loss function. A sharp absorption onset appears near 1.3 eV, followed by strong peaks at ~ 3.2 eV and ~ 4.6 eV, indicating excellent visible-light harvesting. The corresponding plasmonic loss peaks at ~ 4.2 to 4.8 eV confirm strong collective electronic oscillations, supporting high optical activity. Figure 3(d) shows the real and imaginary parts of optical conductivity, where a dominant conductivity rise begins above 2 eV, aligning with the strong absorption region and confirming efficient photo-carrier generation.

These optical properties strongly support the suitability of K2GeI6 for high-efficiency photovoltaic operation under standard solar illumination conditions. The pronounced absorption in the visible region ensures efficient utilization of the solar spectrum, promoting strong photo-carrier generation. The high optical conductivity confirms efficient charge transport and reduced recombination losses, which are critical for achieving high power conversion efficiency. Additionally, the presence of strong plasmonic activity enhances light–matter interaction, potentially improving energy harvesting efficiency in thin-film photovoltaic architectures.

Figure 3(e) illustrates the dielectric function, where the real part peaks around 6.0 at ~4.25 eV, signalling strong polarizability and reduced exciton binding energy-factors that facilitate better charge extraction. Figure 3(f) shows the refractive index, which reaches a maximum value of ~1.75 near 3.5 eV, consistent with dense optical transitions in the visible spectrum.

Finally, Fig. 3(g) presents the reflectivity spectrum, which remains relatively low (<0.1) across the visible range, with moderate peaks at ~2.8 eV and ~5 eV, confirming that K_2GeI_6 allows most incident sunlight to enter rather than being reflected-essential for high-efficiency PV operation. Since K_2GeI_6 contains heavy iodine atoms, spin–orbit coupling (SOC) effects may influence its electronic structure, particularly near the conduction and valence band edges. SOC can contribute to slight bandgap modifications and affect carrier dynamics as well as optical transitions in halide perovskite systems. Although explicit SOC-based calculations were beyond the scope of the present work, its potential influence on the electronic and optical properties of K_2GeI_6 has been acknowledged and discussed. Future studies involving SOC-inclusive DFT calculations may provide deeper insight into relativistic effects and their impact on photovoltaic performance.

3.2 Optimization of proposed solar cell using SCAPS-1D simulations

3.2.1 Role of ETL/HTL selection on device characteristics

Figure S1(a) represents the effect of Voc with different ETL/HTL combinations. WS_2/MoO_3 (1.01 V) and ZnO/MoO_3 (1.00 V) achieve the highest Voc due to favorable band alignment at the interfaces, which minimizes interfacial recombination and enhances carrier separation. In contrast, devices with CuO as the HTL display reduced Voc (0.95–0.96 V), arising from a larger energy offset between the absorber and HTL that promotes non-radiative recombination. This highlights that MoO_3 provides better energetic matching with K_2GeI_6, enabling efficient hole extraction and reduced recombination losses. Figure S1(b) represents the Jsc trends, where WS_2/MoO_3 records the highest photocurrent density (32.41 mA/cm^2), followed closely by ZnO/MoO_3 (32.27 mA/cm^2). This improvement is attributed to efficient charge collection at the ETL/absorber interface, as WS_2 and ZnO both possess suitable conduction band positions and high electron mobility. Conversely, CuO-based devices show lower Jsc (29.23–30.90 mA/cm^2) because of poor hole transport and interfacial mismatch, which increase carrier trapping and recombination. This confirms that MoO_3 enhances carrier balance by providing a smoother hole extraction pathway.

Figure S1(c) depicts the FF performance across devices, where all configurations show high values above 84%, demonstrating good diode quality and reduced resistive losses. WS_2/MoO_3 (85.78%) and ZnO/MoO_3 (85.79%) exhibit the highest FF, reflecting minimized interfacial resistance and efficient charge extraction. In comparison, CuO-based PSCs yield slightly lower FF (84.90–85.48%), indicating increased interfacial resistance and higher non-radiative recombination. This trend further confirms that MoO_3 is superior to CuO as an HTL, as it supports stable and symmetric charge transport. Figure S1(d) shows the overall PCE performance, where WS_2/MoO_3 achieves the maximum efficiency (27.83%), slightly outperforming ZnO/MoO_3 (27.70%). The superior efficiency originates from the simultaneous enhancement of Voc, Jsc, and FF in MoO_3-based devices. In contrast, CuO-based devices exhibit significantly lower efficiencies (23.67–25.26%) due to cumulative losses in Voc and Jsc. This confirms that CuO introduces higher recombination and weaker hole extraction efficiency, making it less suitable for K_2GeI_6 PSCs. Collectively, the results demonstrate that the WS_2/MoO_3 configuration provides the most favorable device physics, with efficient band alignment, superior charge transport, and suppressed recombination.

WS_2 was selected as the ETL due to its high electron mobility, suitable conduction band alignment with K_2GeI_6, and excellent chemical stability, which collectively facilitate efficient electron extraction and reduced recombination losses. Similarly, MoO_3 was employed as the HTL because of its deep work function, favorable valence-band alignment, and efficient hole transport characteristics, enabling improved hole extraction and enhanced device performance. Recent literature reports on WS_2- and MoO_3-based perovskite solar cells have also demonstrated their effectiveness in improving charge transport and suppressing interface recombination.

3.2.2 Band alignment study

Figure 4(a) shows the schematic band alignment of the PSC with FTO/WS_2/K_2GeI_6/MoO_3/Au configuration. The conduction band offset between K_2GeI_6 (1.27 eV) and WS_2 (1.8 eV) is small and favorable, enabling efficient electron extraction while preventing backflow of holes. Similarly, MoO_3 (3.17 eV) aligns well with the valence band of K_2GeI_6, ensuring fast hole transport and minimizing interfacial recombination. The overall architecture demonstrates smooth energy cascades at both ETL and HTL sides, which supports efficient carrier separation and selective extraction. This favorable band alignment is one of the primary reasons behind the enhanced Voc and Jsc observed in WS_2/MoO_3-based PSC.

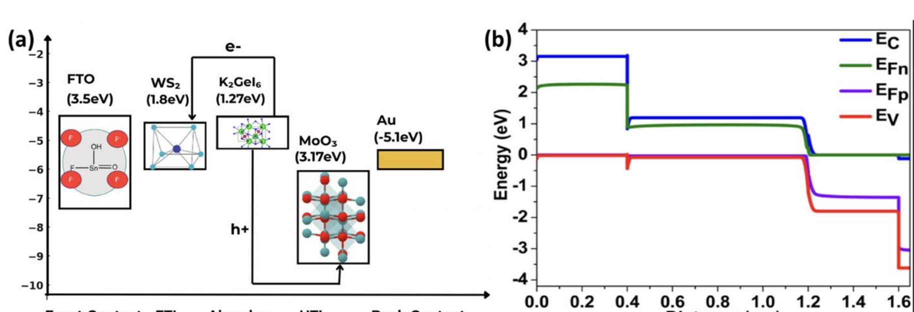

Fig. 4 (**a**) Band alignment and (**b**) SCAPS-1D based band level diagram of our device

Figure 4(b) represents the simulated energy band diagram of the complete device under standard working conditions (AM 1.5G spectrum and 300 K). The conduction band (E_c), valence band (E_v), and quasi-Fermi levels (E_{Fn} for electrons and E_{Fp} for holes) remains separated, which indicates strong carrier selectivity and minimal recombination losses. The flat quasi-Fermi levels inside the absorber region confirm efficient charge transport under illumination. The sharp bending near the interfaces reflects presence of built-in electric fields, which facilitate charge drift toward selective contacts. This alignment ensures high Voc and stable device performance by reducing defect-assisted recombination pathways.

3.2.3 Synergistic effect of acceptor density (N_A) and absorber thickness on solar cell performance

Figure S2(a) shows the dependence of Voc on absorber thickness and N_A. Voc improves with increasing thickness due to enhanced light absorption and reduced surface recombination, saturating beyond ~ 800 nm. At very low N_A, Voc is limited by weak built-in fields, while very high N_A (> 10^{18} cm^{-3}) causes increased Auger recombination, increasing Voc. The optimal range is higher N_A with sufficient absorber thickness, ensuring strong built-in potential and suppressed recombination. Figure S2(b) represents the variation of Jsc. Higher absorber thickness allows more photon absorption, improving Jsc after ~ 600 nm. Beyond this, the gain is marginal due to recombination losses inside the bulk. At low N_A, poor electric field strength reduces carrier collection, leading to lower Jsc. At very high N_A, increased carrier scattering and trap-assisted recombination again suppress Jsc. Thus, a balanced doping density with intermediate absorber thickness maximizes Jsc.

Figure S2(c) depicts the FF trend. FF remains high and stable for moderate N_A across a wide thickness range, reflecting reduced resistive losses and balanced charge transport. At very low N_A, poor electric fields increase series resistance, lowering FF. At very high N_A, enhanced recombination reduces carrier lifetime, again dropping FF. This indicates that both under-doping and over-doping degrade diode quality, while moderate N_A stabilizes FF close to 80–82%. Figure S2(d) shows the combined effect on PCE. Maximum efficiency (~ 26.60%) is observed for absorber thicknesses between 600 and 1400 nm and N_A ~10^{17} cm^{-3}. Too thin absorbers (< 400 nm) fail to capture sufficient light, while very thick ones (> 1200 nm) may suffer from recombination losses. Similarly, very low doping reduces built-in potential and reduces performance.

3.2.4 Cumulative influence of donor density (N_D) and absorber thickness on device performance

Figure S3(a) shows the variation of Voc with absorber thickness and donor density (N_D). Voc decreases drastically as N_D increases beyond 10^{17} cm^{-3} due to enhanced non-radiative recombination at defect sites, which suppresses quasi-Fermi level splitting. For lower N_D values (≤ 10^{16} cm^{-3}), Voc remains nearly constant, confirming that bulk defect passivation is essential to maintain high Voc. Figure S3(b) represents the influence of absorber thickness and N_D on Jsc. At low defect densities, Jsc reaches its maximum (~ 32 mA/cm^2) because carriers are efficiently collected without significant recombination losses. However, at higher N_D (> 10^{17} cm^{-3}), Jsc reduces sharply since bulk defects act as trap-assisted recombination centers, limiting photocurrent extraction.

Figure S3(c) depicts the variation of FF with absorber thickness and N_D. FF decreases significantly for higher ND values ($> 10^{17}$ cm^{-3}) due to increased series resistance and suppressed carrier transport efficiency. At low N_D, FF remains above 80%, confirming the importance of defect-controlled transport dynamics. Figure S3(d) shows the overall impact on PCE, which follows the combined trend of Voc, Jsc, and FF. The highest efficiency (~ 27%) is achieved at thicker absorber thickness (500–1300 nm) and low N_D ($\leq 10^{16}$ cm^{-3}). At higher defect densities, PCE drops drastically due to recombination losses.

3.2.5 Combined effect of interface defect density (N_T) and absorber thickness on photovoltaic performance

Figure S4(a) depicts the dependence of Voc on absorber thickness and interface defect density (N_t). A high N_t ($\geq 10^{14}$ cm^{-3}) leads to a sharp decrease in Voc due to enhanced non-radiative recombination losses at the ETL/absorber and absorber/HTL interfaces, which reduces quasi-Fermi level splitting. For lower N_t (~ 10^{13}–10^{14} cm^{-3}), Voc remains stable around 1.0 V, confirming that proper interface passivation is vital to sustain higher voltages in proposed PSC. Figure S4(b) shows the variation of Jsc with absorber thickness and N_t. At low N_t, Jsc increases steadily with thickness owing to enhanced light absorption and efficient carrier extraction. However, at higher N_t ($\geq 10^{14}$ cm^{-3}), Jsc is significantly suppressed (< 25 mA/cm^2) because recombination losses at the interfaces dominates, limiting the photocurrent despite increased thickness.

Figure S4(c) represents the influence of N_t on the FF. As N_t increases, FF drops due to carrier trapping and transport losses at defect-laden interfaces. At optimal N_t (~ 10^{14} cm^{-3}), FF exceeds 83%, highlighting the importance of defect-controlled transport pathways. This confirms that reducing interfacial trap states minimizes series resistance and recombination-induced distortions in the J–V curve. Figure S4(d) illustrates the overall impact on PCE, which follows the combined trends of Voc, Jsc, and FF. The highest efficiency (~ 26.2%) is obtained at absorber thickness around 1000–1200 nm with $N_t \leq 10^{14}$ cm^{-3}. Beyond this range, PCE declines sharply due to severe recombination losses at interfaces, even when the absorber thickness is large enough to harvest photons effectively.

3.2.6 Influence of transport layer thickness on device performance

Figure S5(a) presents the variation of FF with ETL and HTL thicknesses. The results reveal that FF remains nearly unaffected by thickness changes, consistently around 85.8–85.9%, with only a marginal increase when ETL thickness exceeds 300 nm and HTL thickness exceeds 300 nm. This insensitivity indicates that FF in PSC is largely dominated by defect states and series resistance, rather than charge transport limitations arising from transport layer thickness. Figure S5(b) depicts the corresponding trend in PCE with change in ETL-HTL thickness. Unlike FF, PCE has slight correlation with increasing ETL thickness, reaching a maximum of ~ 28.44% at ETL = 400 nm and HTL ≥ 50 nm. At lower ETL thickness (< 200 nm), PCE drops below 28%, suggesting incomplete charge extraction and higher recombination at thin ETL layers. The weak dependence on HTL thickness confirms that charge transport in the hole channel is efficient even for relatively thin HTLs.

3.2.7 effect of R_S and R_{Sh} resistances

Figure 5(a) shows the influence of series resistance (R_s) on the PV parameters of K_2GeI_6-PSC. A gradual rise in R_s from 1 to 10 Ω·cm² results in a clear decline in PCE (~27 → ~19%) and FF (83 → 57%). Meanwhile, Jsc and Voc stays nearly unchanged. This behavior originates from the fact that higher R_s increases internal resistive losses, obstructing carrier transport and causing reduced current flow and degraded power conversion efficiency.

Figure 5(b) presents the variation of shunt resistance (R_{sh}) and its effect on PV metrics. As R_{sh} increases from 200 to 2000 Ω·cm², both PCE and FF rise significantly, saturating at ~28% and ~84%, respectively, while Jsc and Voc remain stable. This improvement arises because higher R_{sh} minimizes leakage current pathways and suppresses recombination losses, ensuring efficient carrier collection. For the optimized device configuration, the final resistance parameters employed in the SCAPS-1D simulation were R_s = 1 Ω·cm² and Rsh = 2000 Ω·cm². These values represent standard operating conditions, similar resistance ranges have been experimentally approached in high-quality PSCs through optimized interface engineering, defect passivation, and improved contact formation. Therefore, the selected values were considered appropriate for evaluating the theoretical upper-limit performance of the proposed K_2GeI_6-based device.

3.2.8 Temperature dependence of device performance

Figure 6(a) presents the effect of temperature on PCE and FF. As the temperature increases from 300 K to 400 K, both metrics exhibit a consistent decline, with PCE dropping from ~28.5% to ~23.5% and FF from ~86% to ~81.4%. This reduction arises from thermal agitation and thermally activated defect states, which deteriorate the charge transport pathways at elevated temperatures. Figure 6(b) depicts the variation of Jsc and Voc with temperature. Interestingly, Jsc shows only a marginal increase up to 33.1 mA/cm^2, while Voc decreases slightly till 0.98 V. The slight rise in Jsc can be attributed to enhanced thermal excitation of carriers, which improves mobility and increases the probability of charge collection at the electrodes. In contrast, Voc decreases with rising temperature due to a combined impact of higher intrinsic carrier concentration and bandgap narrowing at elevated thermal conditions. These effects reduce the quasi-Fermi level splitting, thereby lowering Voc [20].

Figure 6(c) shows the quantum efficiency (QE) spectra across a temperature range of 300–400 K. At room temperature (300 K), QE achieves almost 100% in the visible region, confirming efficient light absorption and charge generation. However, as the temperature rises, QE gradually declines, with more pronounced losses in the longer-wavelength (near-IR) region. This reduction can be explained by enhanced non-radiative

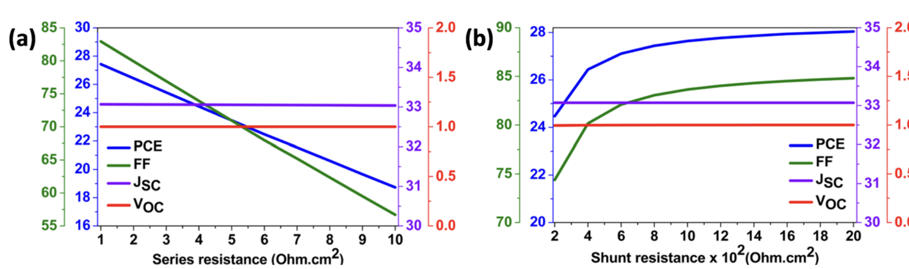

Fig. 5 Variation of (**a**) R_S and (**b**) R_{Sh} and their effect PV metrics

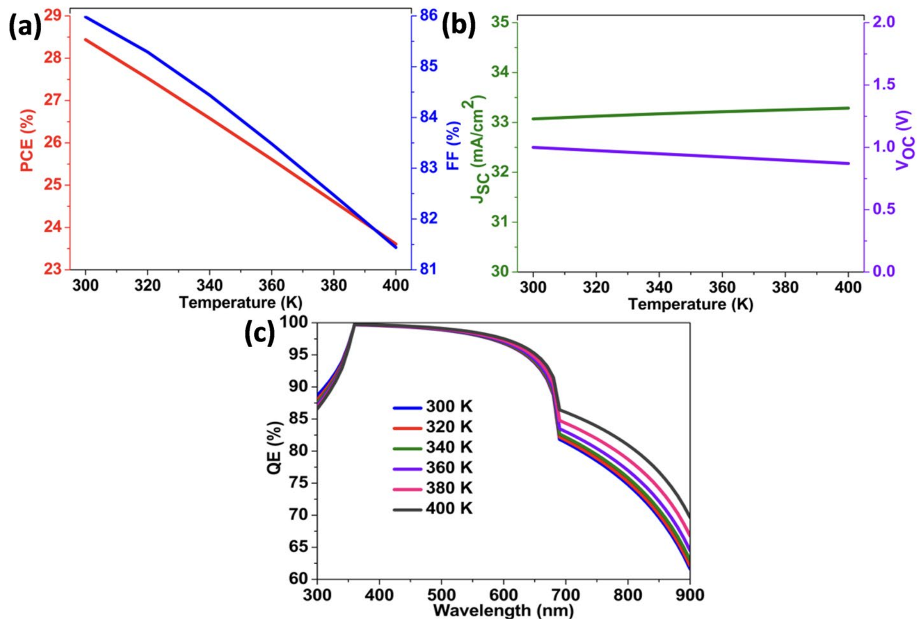

Fig. 6 Variation of temperature and its effect on (**a**) PCE-FF, (**b**) Jsc-Voc and (**c**) QE

recombination and trap-assisted processes that dominate at elevated temperatures. Moreover, thermal vibrations disrupt charge transport, decreasing carrier lifetime and collection efficiency. The combined effect leads to lower spectral response and diminished photon-to-electron conversion capability. This decline in QE further supports the observed drop in PCE and FF, demonstrating the slight sensitivity of K_2GeI_6-based PSCs to thermal stress.

3.3 ML study- model analysis and effect of interface defect density

Figure 7(a) presents the correlation heatmap among the key PV input parameters such as bandgap energy (E_g), electron affinity (χ), dielectric constant (ε_r), electron mobility (μ_e), hole mobility (μ_h), defect density (N_t), and PCE. A strong negative correlation between N_t and PCE (−0.69) emphasizes the detrimental role of trap states, which facilitate non-radiative recombination and limit carrier lifetime. Conversely, mobility parameters (μ_e, μ_h) show positive correlation with efficiency, highlighting that enhanced charge transport is beneficial for minimizing series resistance and maximizing current extraction. This correlation matrix guided the prioritization of parameters fed into the ML framework for defect-sensitive performance prediction.

Figure 7(b) shows the ML-predicted PCE against the actual PCE values. The close alignment along the diagonal (R^2 = 0.976, RMSE = 0.27, MAE = 0.2) demonstrates the robustness and accuracy of the ML model in learning the complex nonlinear mapping between device physics inputs and output efficiency. This validates that the trained model can serve as a fast predictive tool, drastically reducing the need for exhaustive simulations.

In this work, a ML model was trained on a dataset comprising defect-related parameter sets and corresponding PV metrics collected from similar perovskite materials. The model learned the complex correlation between defect states and device performance, and was subsequently used to predict optimized defect parameters for K_2GeI_6. Table 1 lists these ML-extracted parameters-including defect density, capture cross-sections,

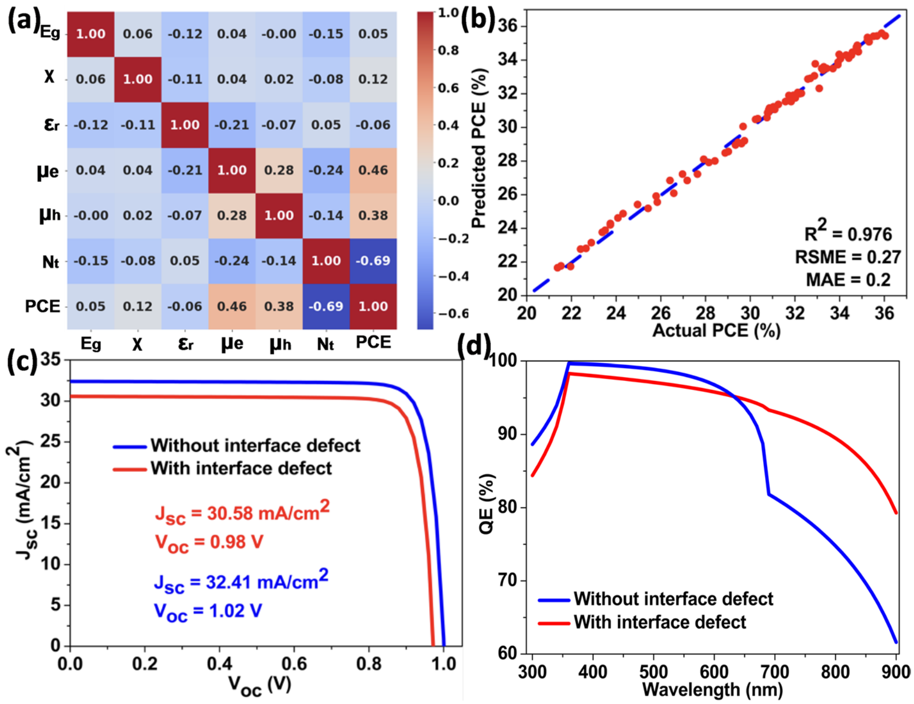

Fig. 7 (**a**) Heatmap for multiple parameters, (**b**) predicted vs. actual PCE, (**c**) JV and (**d**) QE curve of PSC

and trap energy levels-which are otherwise difficult to determine experimentally. The predicted parameters were then embedded into SCAPS-1D simulations to evaluate PV metrics, thereby validating their physical relevance and performance consistency. This integrated ML–SCAPS approach effectively bridges the gap between microscopic defect estimation and macroscopic device performance analysis, enabling a more realistic assessment of defect-induced PV losses.

Figure 7(c) shows the J–V curves of the PSC with and without ML-derived interface defects. In the defect-free configuration, the device achieves higher Jsc and Voc, along with improved FF and overall PCE. When defect parameters from Table 1 are introduced into SCAPS-1D, a noticeable reduction is observed across all PV metrics. The origin lies in trap-assisted recombination at the interfaces: photogenerated carriers are captured by defect states and lost before reaching the electrodes. This suppresses quasi-Fermi level splitting (leading to lower Voc) and reduces the flow of extractable carriers (lower Jsc), while also degrading the fill factor due to recombination-related current losses. Hence, Fig. 7(c) highlights how microscopic defect states directly translate into macroscopic performance deterioration. Upon inclusion of ML-derived interface defects, the photovoltaic parameters exhibit only moderate degradation, with Jsc decreasing by ~5.6%, Voc by ~3.9%, and the overall device performance by ~5–7%, indicating appreciable defect tolerance and reasonable operational stability of the K_2GeI_6 absorber.

Figure 7(d) illustrates the QE spectra under the two conditions-with and without interface defects. For the defect-free case, QE remains above 90% throughout the visible range, confirming efficient absorption and charge transport but sharply decrease after 700 nm. Once defects are included, QE is declined, but does not fall sharply especially at 700 nm and above wavelengths because deeper-generated carriers are already limited by recombination, leading to a smoother decline. In contrast, for the defect-free

Table 1 Input parameters for defect for SCAPS 1D extracted by ML

Interface	Defect type	Capture cross-section (electrons) σ_n (cm^2)	Capture cross-section (holes) σ_p (cm^2)	Energetic distribution	Reference for defect energy level	Energy with respect to reference (eV)	Total density (integrated over energies) (cm^{-2})
WS_2 / K_2GeI_6 (ETL/absorber)	Neutral	1.0×10^{-18}	1.0×10^{-18}	Single	above the highest E_v	0.60	5.0×10^{10}
K_2GeI_6 / MoO_3 (absorber/HTL)	Neutral	1.0×10^{-19}	1.0×10^{-18}	Single	above the highest E_v	0.60	1.0×10^{10}

Table 2 Comparison of our proposed solar cell with other work

Device Configuration	PCE (%)	FF (%)	Jsc (mA/cm^2)	Voc(V)	References
$FTO/WS_2/K_2GeI_6/MoO_3/Au$	27.83	85.78	30.58	0.98	This Paper
$ITO/WS_2/CsSnI_3/CuI/Au$	13.59	73.08	9.26	0.93	[21]
$ITO/WS_2/ MAPbI_3/P_3HT$	27.02	85.44	21.56	1.46	[22]
$FTO/WS_2/CsPbI_3/rGO$	19.85	84.28	21.19	1.11	[23]
$FTO/TiO_2/CsSnI_3/WSe_2$	20.18	84.39	21.76	1.09	[23]
$FTO/TiO_2/CsSnI_3$/Spiro-OMeTAD	27.22	81.74	29.03	1.14	[23]
$FTO/TiO_2/Cs_2SnI_6/Cu_2O$	17.77	80.50	30.00	0.74	[24]
$FTO/TiO_2/Cs_2SnI_6/rGO/Ag$	19.07	83.00	24.36	0.94	[25]
$FTO/ZnO/Cs_2SnI_6/P_3HT/Au$	23.00	77.5	27.09	1.08	[26]
$ITO/CuI/Cs_2SnI_6/ZnO/AzO/Ag$	14.65	71.08	23.91	0.86	[27]
$FTO/MoO_3/A_2LiInBr_6/WS_2/Cu$	17.29	78.26	16.64	1.31	[28]
$ITO/ WS_2/K_2CuCrCl_6/CBTS$	27.99	73.50	31.85	1.2	[29]
$ITO/CeO_2/KSnI_3/CBTS/Ag$	13.46	86.05	21.5	0.86	[30]
$FTO/WSe_2/A_2LiInBr_6/WS_2/Cu$	17.27	78.27	16.72	1.31	[28]

case, carriers travel longer distances without significant loss, so the collection efficiency drops more sharply at longer wavelengths due to intrinsic absorption limits rather than defect trapping. Additionally, defect-induced non-radiative pathways reduce photon-to-electron conversion probability across the visible spectrum. This makes QE a sensitive fingerprint of interfacial defect density, clearly linking microscopic traps with optical response. Table 2 compares the performance of our proposed $FTO/WS_2/K_2GeI_6/MoO_3/Au$ PSC with previously reported devices. It is evident that our design achieves one of the highest PCE values (27.83%) with excellent FF and Jsc. A better and smooth band alignment makes flow of electrons and holes smooth and thus reducing losses contributes increasing PCE values. A perfect optimization of series and shunt resistance gives higher Jsc and Voc and thus in return increases FF.

4 Device feasibilty-experimnetal pathway

The fabrication of the proposed $FTO/WS_2/K_2GeI_6/MoO_3/Au$ device can be carried out through a carefully designed, defect-minimized process to achieve high-quality films and stable interfaces. The procedure may begin with thorough cleaning of FTO substrates using acetone, isopropanol, and deionized water, followed by UV–ozone treatment to remove organic contaminants and enhance surface energy. A thin WS_2 (ETL) can then be deposited via spin coating or sputtering, forming a compact and uniform film after annealing at moderate temperatures (150–200 °C) under nitrogen. This WS_2 layer is expected to facilitate efficient electron extraction while suppressing interface-induced recombination [31]. The K_2GeI_6 absorber layer can be synthesized by a two-step method, where a GeI_2 precursor film is first spin-coated and subsequently converted to K_2GeI_6 through reaction with a potassium iodide solution [32]. Controlled annealing at 100–140 °C in an inert environment can promote uniform crystallization and minimize halide vacancies or non-stoichiometric defects. Additionally, incorporating a slight excess of KI or small Lewis-base additives in the precursor can further passivate deep-level traps and improve grain boundary quality. Following the absorber formation, a thin MoO_3 layer can be thermally evaporated as the HTL, ensuring proper energy-level alignment and efficient hole extraction. A surface treatment using phenethylammonium iodide (PEAI) or similar passivators may be applied prior to MoO_3 deposition to reduce interfacial trap states and prevent iodine migration [33]. Finally, a gold layer can be thermally evaporated under high vacuum to form the back electrode, completing the device [34]. The efficiency can be increased with help of anti-reflective coating which will prevent reflection of sunlight [35]. Through such a controlled, multistep approach-integrating defect passivation, optimized annealing, and interfacial engineering the fabrication process can yield stable and efficient K_2GeI_6-based perovskite solar cells. Combined with machine learning-guided defect modeling and SCAPS-1D validation, this strategy offers a novel, scalable route for producing defect-tolerant, lead-free PV device.

Though the above section presents a proposed conceptual fabrication pathway intended to provide a feasible experimental roadmap for realizing the simulated K_2GeI_6-based solar cell architecture, rather than an experimentally validated protocol. The suggested deposition and interface-engineering approaches are based on established thin-film fabrication techniques reported for related halide perovskite systems. However, practical challenges such as achieving phase-pure K_2GeI_6 films, controlling defect formation, maintaining stoichiometric composition, and minimizing processing-induced instabilities may significantly influence experimental device realization and performance. Therefore, further experimental studies are required to validate and optimize the proposed fabrication methodology.

5 Conclusion

In this work, a comprehensive multiscale computational framework combining Density Functional Theory (DFT), SCAPS-1D simulation, and machine learning-assisted defect analysis was employed to investigate the photovoltaic potential of lead-free K_2GeI_6-based perovskite solar cells. The optimized FTO/WS_2/K_2GeI_6/MoO_3/Au device demonstrated excellent photovoltaic performance with a power conversion efficiency (PCE) of 27.83%, fill factor (FF) of 85.78%, short-circuit current density (Jsc) of 30.58 mA/cm^2, and open-circuit voltage (Voc) of 0.98 V. DFT calculations confirmed favorable electronic and optical characteristics of K_2GeI_6, including a suitable direct bandgap, strong visible-light absorption, and promising charge-transport behavior. SCAPS-1D simulations further revealed the critical influence of defect density, transport-layer selection, absorber thickness, resistance effects, and temperature variation on device performance. In particular, the WS_2/MoO_3 transport-layer combination provided favorable band alignment and efficient charge extraction, contributing to suppressed recombination losses and enhanced photovoltaic output. The machine learning-assisted defect engineering approach enabled identification of dominant defect-sensitive parameters and provided predictive insight into defect-governed photovoltaic behavior. Although the introduction of interface defects caused moderate performance degradation, the device retained comparatively stable operation, indicating appreciable defect tolerance of K_2GeI_6-based architectures. Compared with several previously reported lead-free and Ge/Sn-based perovskite solar cells, the proposed device demonstrates competitive photovoltaic performance along with a systematic defect-sensitive optimization strategy. While the present work remains computational in nature and requires further experimental validation, the integrated DFT–SCAPS–ML methodology established here provides a promising framework for the design and optimization of stable, lead-free, and defect-tolerant next-generation perovskite photovoltaic devices.

Supplementary Information

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Supplementary material 1.

Author contributions

Manasvi Raj: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. Kaushiki Chhabra: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis. Advay Makhija: Writing – original draft, Software, Resources, Methodology. Gorang Singh: Writing – original draft, Software. Neeraj Goel: Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

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Data availability

The data that support the findings of this study are available on request from the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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